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MT6685 Clock IC Datasheet

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Version History

Version	Date	Author	Description
0.1	2021-05-05	Yunghsu Chen	Initial draft
0.2	2021-05-21	Yunghsu Chen	Modified AUXADC power and ground pins description.
0.3	2021-07-13	John Chiang	Modified RTC accuracy and duty cycle.
0.4	2021-08-20	John Chiang	1. Modified RFCK1 10 kHz offset phase noise 2. Modified BBCK output impedance mapping table
0.5	2021-08-25	Chuck Chiu	Modified power on/off sequence.
0.6	2021-11-05	John Chiang	1. Added 26 MHz clk phase noise spec. for RFCK1. 2. Added 52 MHz clk phase noise spec. for RFCK2. 3. Modified RG_XO_EXTBUF_BBCK1_RSEL code description.
0.7	2021-11-29	Yunghsu Chen	Updated POD.
0.8	2022-01-04	Yunghsu Chen Joseph Cheng	1. Updated top marking. 2. Update BBCK_HD description.
0.9	2022-03-24	Yunghsu Chen	1. Updated thermal information. 2. Removed VRTC Vin min. voltage and modified UVLO/DDLO description.
1.0	2022-04-11	Yunghsu Chen Katy Chang Chuck Chiou	1. First official release. 2. Updated die size information in Figure 1-1. 3. Updated DCXO description in 3.2.5. 4. Updated sequence figure. 5. Updated table in 2.5 (FAULTB, EN, RTC_INT).
1.1	2022-06-13	Chuck Chiou Joseph Cheng John Chiang	1. Updated VRTC power on waveform in Figure 3-2. 2. Added debug status register. 3. Added 10 Hz/100 Hz phase noise spec. for RFCK2/BBCK.
1.2	2022-09-22	Yunghsu Chen	Updated top marking.
1.3	2023-03-13	Yunghsu Chen	1. Updated power on sequence note. 2. Updated operating current electrical characteristic in Table 2-4.
1.4	2023-09-06	Yunghsu Chen	Updated descriptions of Figure 3-2. MT6685HP/B power-on/off control sequence without XTAL by EN high.

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1 Overview

1.1 Features

- Handles 5G-Module baseband clock IC
- Input range: 2.6~5V
- 9 clock buffers and 4 input channels for ADC optimized for specific 5G-Module subsystems
- 32K-Crystal-less RTC oscillator for system timing, 1.8V clock buffer output
- SPMI interface
- Programmable under voltage lockout protection
- Watchdog reset
- Flexibility hardware clock IC reset function
- Power-on reset and start-up timer
- 42-pin WLCSP package

1.2 Applications

MT6685 is ideal for clock IC of 5G-Module and other portable systems.

1.3 General Descriptions

MT6685 is a clock IC system chip optimized for 5G-Module, containing nine clock buffers and four input channels for ADC optimized for specific 5G-Module subsystems.

Sophisticated controls are available for power-up and the RTC alarm. MT6685 is optimized for maximum battery life, allowing the RTC circuit to stay alive without a battery for several hours.

MT6685 adopts SPMI interface and one SRCLKEN control pins to control DCXO, LDOs, and various drivers; it provides enhanced safety control and protocol for handshaking with baseband.

MT6685 is available in a 42-pin WLCSP package. The operating temperature ranges from -30°C to +85°C.

1.4 Ordering Information

Order #	Marking	Temp. range	Package
MT6685HP/B		-30 ~ +85°C	WLCSP 42 pins
MT6685LP/B		-30 ~ +85°C	WLCSP 42 pins

Note: H for 52M XTAL; L for 26M XTAL.

1.5 Top Marking Definition

MT6685HP/B

MT6685LP/B



YWW: Date Code

\$: Random Code

Pin assignments and descriptions:

42	1	2	3	4	5	6	
A	VAUX18	VRTC	AUXADC_VIN4	AUXADC_VIN2	RTC_INT	SDA	A
B	AVSS18_AUXADC	AUXADC_VIN5	AUXADC_VIN3	FAULT_B	RTC32K_1V8_F	SCL	B
C	VREF	AVSS_VREF	EN	WDTRSTB_IN	RTC32K_1V8_0	SRCLKENA0	C
D	DVDD18_DIG	VBAT	RFCK1B	RFCK2B	BBCK2	FSOURCE	D
E	AUXADC_VIN1	DVDD18_IO	RFCK1A	RFCK2A	BBCK1	BBCK5	E
F	XTAL2	AVSS_XO	AVSS_RFCK1	AVSS_RFCK2	DGND	BBCK4	F
G	XTAL1	VXO	VRFCCK1	VRFCCK2	VBBCK	BBCK3	G
	1	2	3	4	5	6	

Figure 1-1. MT6685 WLCSP 42 (2.2306*2.4958 mm) pin assignment (top view)

Table 1-1. MT6685 pin description

Ball	Symbol	I/O	Description
A1	VAUX18	PWR	AUXADC power
A2	VRTC	PWR	RTC LDO output. Supply of RTC macro where backup battery can be added.
A3	AUXADC_VIN4	I	AUXADC input
A4	AUXADC_VIN2	I	AUXADC input
A5	RTC_INT	O	RTC interrupt
A6	SDA	I/O	SPMI control interface
B1	AVSS18_AUXADC	GND	AUXADC ground
B2	AUXADC_VIN5	I	External resistor
B3	AUXADC_VIN3	I	AUXADC input
B4	FAULT_B	O	FAULTB indicator

Ball	Symbol	I/O	Description
B5	RTC32K_1V8_F	O	VIO18 domain 32 kHz clock output
B6	SCL	I	SPMI control interface
C1	VREF	O	Bandgap reference voltage
C2	AVSS_VREF	GND	VDIG18 output voltage
C3	EN	I	Enable control, active high (The operating voltage should be $\leq$ VSYS.)
C4	WDTRSTB_IN	I	Watchdog reset from AP
C5	RTC32K_1V8_0	O	VIO18 domain 32 kHz clock output
C6	SRCLKENA0	I	Source clock enable pin 0
D1	DVDD18_DIG	PWR	Digital power
D2	VBAT	PWR	VSYS supply input for internal block and UVLO detection
D3	RFCK1B	O	RFCK clock output
D4	RFCK2B	O	RFCK clock output
D5	BBCK2	O	BBCK clock output
D6	FSOURCE	PWR	eFuse power source
E1	AUXADC_VIN1	I	AUXADC input
E2	DVDD18_IO	PWR	Digital I/O power
E3	RFCK1A	O	RFCK clock output
E4	RFCK2A	O	RFCK clock output
E5	BBCK1	O	BBCK clock output
E6	BBCK5	O	BBCK clock output
F1	XTAL2	I/O	XTAL input 2
F2	AVSS_XO	GND	Ground
F3	AVSS_RFCK1	GND	Ground for RF clock buffer
F4	AVSS_RFCK2	GND	Ground for RF clock buffer
F5	DGND	GND	Ground
F6	BBCK4	O	BBCK clock output
G1	XTAL1	I/O	XTAL input 1
G2	VXO	PWR	Power supply of XO
G3	VRFC1	PWR	Power supply of XO buffer
G4	VRFC2	PWR	Power supply of XO buffer
G5	VBBCK	PWR	Power supply of XO buffer
G6	BBCK3	O	BBCK clock output

2 Electrical Characteristics

2.1 Absolute Maximum Ratings over Operating Free-Air Temperature Range

Stresses beyond those listed in [Table 2-1](#) might cause permanent damage to the device. These numbers are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect the device reliability.

Table 2-1. Absolute maximum ratings

Parameter	Condition	Min.	Typical	Max.	Unit
Free-air temperature range		-40		85	°C
Storage temperature range		-65		150	°C
Battery pin input ⁽¹⁾	Steady state	-0.5		6	V
	Transient (< 10 ms)	-0.5		7	V
Non-battery power pin ⁽²⁾	Steady state	-0.5		3	V
Signal pins ⁽³⁾	Steady state	-0.5		V _{xx} + 0.5 ⁽³⁾	V
ESD robustness	HBM	2,000			V

Note:

1. VBAT -> battery input pin

2. Non-battery power input -> reference [Table 1-1](#) (PWR pin but not connected with battery)

3. V_{xx} = Max. operation voltage (refer to [Table 2-2](#))

2.2 Thermal Characteristics

Parameter	Condition	Min.	Typical	Max.	Unit
Thermal resistance from junction to ambient	In free air		70.5		°C/W

Note: The device is mounted on a 4-metal-layer PCB and modeled per JEDEC51-7 condition.

2.3 Pin Voltage Range

The table below lists the operation range voltages for all MT6685 I/O pins.

Table 2-2. Pin voltage range

Symbol	Voltage range	Unit
VAUX18	0~1.98	V
VRTC	0~3.1	V
AUXADC_VIN4	0~1.98	V
AUXADC_VIN2	0~1.98	V
RTC_INT	0~5	V
SDA	0~1.98	V
AVSS18_AUXADC	0	V
AUXADC_VIN5	0~1.98	V

Symbol	Voltage range	Unit
AUXADC_VIN3	0~1.98	V
FAULT_B	0~5	V
RTC32K_1V8_F	0~1.98	V
SCL	0~1.98	V
VREF	0~1.32	V
AVSS_VREF	0	V
EN	0~5	V
WDTRSTB_IN	0~1.98	V
RTC32K_1V8_0	0~1.98	V
SRCLKENA0	0~1.98	V
DVDD18_DIG	0~1.98	V
VBAT	0~5	V
RFCK1B	0~1.54	V
RFCK2B	0~1.54	V
BBCK2	0~1.32	V
FSOURCE	0~5	V
AUXADC_VIN1	0~1.98	V
DVDD18_IO	0~1.98	V
RFCK1A	0~1.54	V
RFCK2A	0~1.54	V
BBCK1	0~1.32	V
BBCK5	0~1.32	V
XTAL2	0~1.32	V
AVSS_XO	0	V
AVSS_RFCK1	0	V
AVSS_RFCK2	0	V
DGND	0	V
BBCK4	0~1.32	V
XTAL1	0~1.98	V
VXO	0~1.76	V
VRCK1	0~1.54	V
VRCK2	0~1.54	V
VBBCK	0~1.32	V
BBCK3	0~1.32	V

2.4 Recommended Operating Range

Table 2-3. Operation condition

Parameter	Condition	Min.	Typical	Max.	Unit
Ambient temperature (TA)		-30		85	°C
Junction temperature (TJ)		-30		125	°C
Operating input voltage		3.05 (Note)		5	V

Note: This minimum input voltage still needs to check the detailed test conditions for each function in specification table.

2.5 Electrical Characteristics

- VBAT = 2.6~5V, minimum loads applied on all outputs, unless otherwise noted.
- Typical values are at TA = 25°C.

Table 2-4. General electrical specifications

Parameter	Condition	Min.	Typical	Max.	Unit
Operation ground current					
Normal mode ¹	VBAT = 4V, normal mode		307		uA
	VIO18 = 1.8V, normal mode		18		mA
Standby mode ²	VBAT = 4V, low-power mode		66		μA
	VIO18 = 1.8V, low-power mode		277		μA
Power-down leakage current mode ³	VBAT = 4V, off mode Temp = 25°C			88	uA
Under voltage lock-out (UVLO)					
Under voltage falling threshold		2.45	2.5	2.55	V
Under voltage rising threshold		2.75	2.8	2.85	V
EN					
High voltage		0.9			V
Low voltage				0.4	V
Pull low resistor			2M		Ω
RTC_INT					
Output high		VSYS-0.4			V
Output low				0.2	V
Control input voltage					
Control input high (SPMI)		0.7*VIO			V
Control input low (SPMI)				0.3*VIO	V
Control input high (SRCLKEN related)		0.75*VIO			V
Control input low (SRCLKEN related)				0.25*VIO	V

1. Normal mode is the supply current from a main battery and VIO18 separately with MT6685W on. Clock with 20 pF load and regulator outputs are unloaded. See Table 2-5 for more information.
2. Standby mode is the supply current from a main battery and VIO18 separately with MT6685W in standby mode. Clock with 20 pF load and regulator outputs are unloaded. See Table 2-6 for more information.
3. Power-down leakage is the total supply current from a main battery with MT6685 off.

Table 2-5. Normal mode and voltage supplies

Resource	Mode	Voltage supply
Regulator		
VRTC	ON	VBAT
DVDD18_DIG	ON	VBAT
VAUX18	ON	VBAT
VXO	ON	VIO18
VRFC1	ON	VIO18
VRFC2	ON	VIO18
VBBCK	ON	VIO18
Infrastructure		
VREF	ON	VBAT
Clocking		
RFCK1A	ON	VIO18
RFCK1B	ON	VIO18
RFCK2A	ON	VIO18
RFCK2B	ON	VIO18
BBCK1	ON	VIO18
BBCK2	ON	VIO18
BBCK3	ON	VIO18
BBCK4	ON	VIO18
BBCK5	ON	VIO18
RTC32K_1V8_0	ON	VIO18
RTC32K_1V8_F	ON	VIO18

Table 2-6. Standby mode and voltage supplies

Resource	Mode	Voltage supply
Regulator		
VRTC	ON	VBAT
DVDD18_DIG	ON	VBAT
VAUX18	OFF	VBAT
VXO	ON	VBAT
VRFC1	OFF	VIO18
VRFC2	OFF	VIO18
VBBCK	OFF	VIO18
Infrastructure		
VREF	ON	VBAT
Clocking		
RFCK1A	OFF	VIO18
RFCK1B	OFF	VIO18
RFCK2A	OFF	VIO18
RFCK2B	OFF	VIO18
BBCK1	OFF	VIO18
BBCK2	OFF	VIO18
BBCK3	OFF	VIO18
BBCK4	OFF	VIO18
BBCK5	OFF	VIO18
RTC32K_1V8_0	ON	VIO18

Resource	Mode	Voltage supply
RTC32K_1V8_F	ON	VIO18

Table 2-7. Off mode and voltage supplies

Resource	Mode	Voltage supply
Regulator		
VRTC	ON	VBAT
DVDD18_DIG	ON	VBAT
VAUX18	OFF	VBAT
VXO	OFF	VBAT
VRFC1	OFF	VIO18
VRFC2	OFF	VIO18
VBBCK	OFF	VIO18
Infrastructure		
VREF	ON	VBAT
Clocking		
RFCK1A	OFF	VIO18
RFCK1B	OFF	VIO18
RFCK2A	OFF	VIO18
RFCK2B	OFF	VIO18
BBCK1	OFF	VIO18
BBCK2	OFF	VIO18
BBCK3	OFF	VIO18
BBCK4	OFF	VIO18
BBCK5	OFF	VIO18
RTC32K_1V8_0	OFF	VIO18
RTC32K_1V8_F	OFF	VIO18

2.6 Regulator Output

2.6.1 Regulator Output (LDO)

Table 2-6. VAUX18 specifications

Parameter	Condition	Min.	Typical	Max.	Unit
V _{in} range		3	-	5	V
V _{out}		-	1.84	-	V
C _{out}		-	1	-	uF
Turn-on rise time	To within 1% of final voltage	-	-	250	us
Quiescent current		-	52	-	uA
Power-down current	V _{EN} = GND, shutdown	-	-	0.1	uA

Table 2-7. DVDD18_DIG specifications

Parameter	Condition	Min.	Typical	Max.	Unit
V _{in} range		2.5	4	5	V
V _{out}		-	1.8	-	V
C _{out}		-	1	-	uF
Turn-on rise time	To within 1% of final voltage	-	-	300	us
Quiescent current		-	6	-	uA
Power-down current	V _{EN} = GND, shutdown	-	-	0.1	uA

Table 2-8. VXO_LDO specifications

Parameter	Condition	Min.	Typical	Max.	Unit
V _{in} range			1.8		V
V _{out}		-	1.6	-	V
C _{out}		-	1	-	uF
Turn-on rise time	To within 1% of final voltage	-	-	300	us
Quiescent current		-	300	-	uA
Power-down current	V _{EN} = GND, shutdown	-	-	0.1	uA

Table 2-9. VRFCK1/VRFCK2_LDO specifications

Parameter	Condition	Min.	Typical	Max.	Unit
V _{in} range			1.8		V
V _{out}		-	1.4	-	V
C _{out}		-	1	-	uF
Turn-on rise time	To within 1% of final voltage	-	-	300	us
Quiescent current		-	255	-	uA
Power-down current	V _{EN} = GND, shutdown	-	-	0.1	uA

Table 2-10. VBBCK_LDO specifications

Parameter	Condition	Min.	Typical	Max.	Unit
V _{in} range			1.8		V
V _{out}		-	1.2	-	V
C _{out}		-	1	-	uF
Turn-on rise time	To within 1% of final voltage	-	-	300	us
Quiescent current		-	75	-	uA
Power-down current	V _{EN} = GND, shutdown	-	-	0.1	uA

Table 2-11. VRTC_LDO specifications

Parameter	Condition	Min.	Typical	Max.	Unit
V _{in} range		-	4	5	V
V _{out}		-	2.8	-	V
C _{out}		-	22	-	uF
Turn-on rise time	To within 1% of final voltage	-	-	10	mS
Quiescent current		-	3.5	-	uA
Power-down current	V _{EN} = GND, shutdown	-	-	0.5	uA

2.7 AUXADC

Table 2-12. XOADC specifications

Parameter	Condition	Min.	Typical	Max.	Unit
ADC reference voltage		-	1.84	-	V
ADC resolution (LSB)		-	56.15	-	uV
ADC absolute accuracy		-	-	17.54	mV
AUXADC_VIN1-5 voltage range	Guaranteed	0.04	-	1.78	V
AUXADC_VIN1-5 input impedance		10M			
100 K pull-up error	Trimmed value	-	±1	-	%
400 K pull-up error	Trimmed value	-	±1	-	%
30 K pull-up error	Trimmed value	-	±1	-	%
Pull-up temperature coefficient		-	±200	-	ppm/°C

2.8 DCXO

This clock IC integrates a core 26 MHz or 52 MHz XTAL oscillator (XO) and provides clock outputs for higher-level handsets. The output clocks are divided into three groups.

- Low noise clock output for RF communication: RFCK1A and RFCK1B
- Low noise clock output supporting divided-by-2 for RF communication: RFCK2A and RFCK2B
- Low power clock output for baseband circuits: BBCK1~5

Table 2-13. XO specifications

Parameter	Condition	Title	Min.	Typical	Max.	Unit
Frequency	Set by external crystal	F _{dcxo}		26/52		MHz
Startup time	VXO ready to Δfreq. < 1 ppm	T _{startup}			3	ms
Supply voltage = VXO				1.6		V
Supply current At LPM At FPM	Measured form VXO			30/60 1.2/2.4		uA mA
Accuracy At FPM	With calibrated CAPID		-5		5	ppm
Accuracy At LPM			-100		100	ppm

The buffer mode is supported by driving AC input from the XTAL1 pin. The input buffer clock requirements are listed below.

Table 2-14. Reference input buffer specifications (XTAL1, buffer mode)

Parameter	Condition	Min.	Nom.	Max.	Unit
Input impedance - Rp	Rp Cp @26 MHz	30			kΩ
Input impedance - Cp	Rp Cp @26 MHz		2	3	pF

2.8.1 Reference Output Buffer(s) Specifications

Table 2-15. RF clock output buffer specifications (RFCK1A, RFCK1B, 52 MHz)

Test conditions: TSX = 52 MHz, 20 pF capacitive load for each buffer output

Parameter	Condition	Min.	Typical	Max.	Unit
Output frequency			52		MHz
Output swing	Set by VRCK1	1.35	1.4	1.45	V
Duty cycle		47		53	%
Output impedance	1X	40	50	60	Ω
	2X	25	30	35	Ω
	3X	20	25	30	Ω
	4X	12	15	18	Ω
Phase noise	At 10 Hz		-77		dBc/Hz
	At 100 Hz		-107		dBc/Hz
	At 1 kHz		-136		dBc/Hz
	At 10 kHz		-153		dBc/Hz
	At 100 kHz		-160.5		dBc/Hz
	At 1 MHz		-161		dBc/Hz
Supply current of each clock	Measured from VRCK1		2.2		mA

RFCK1 measurement results:

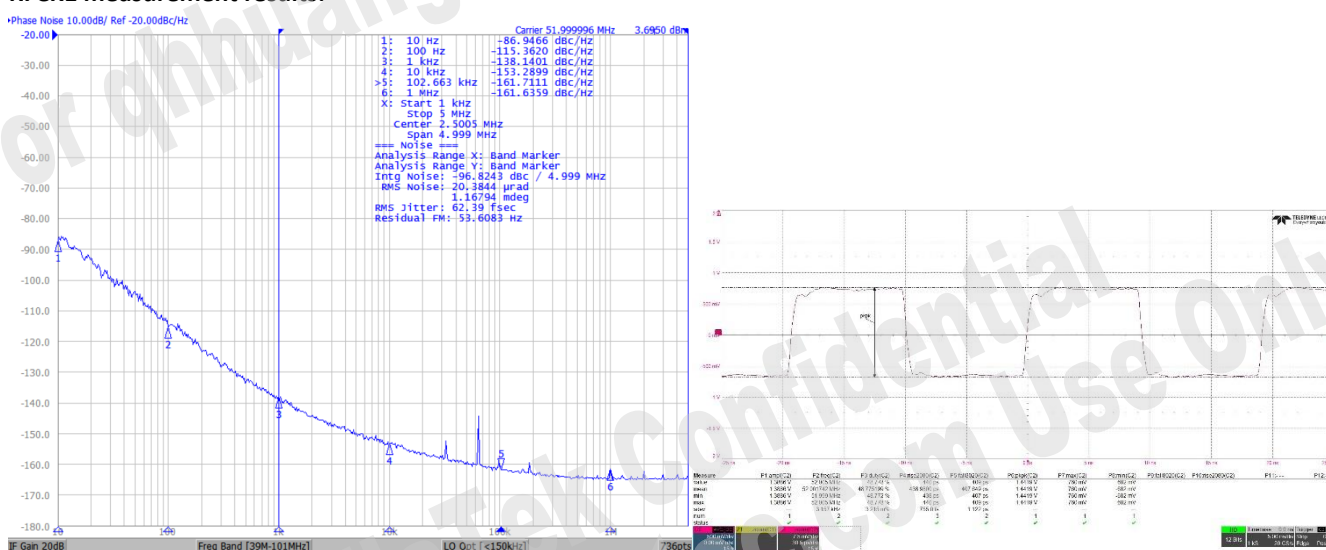


Table 2-16. RF clock output buffer specifications (RFCK1A, RFCK1B, 26 MHz)

Test conditions: TSX = 26 MHz, 20 pF capacitive load for each buffer output

Parameter	Condition	Min.	Typical	Max.	Unit
Output frequency			26		MHz
Output swing	Set by VRFCK1	1.35	1.4	1.45	V
Duty cycle		47		53	%
Output impedance	1X	40	50	60	Ω
	2X	25	30	35	Ω
	3X	20	25	30	Ω
	4X	12	15	18	Ω
Phase noise	At 10 Hz		-80		dBc/Hz
	At 100 Hz		-110		dBc/Hz
	At 1 kHz		-136		dBc/Hz
	At 10 kHz		-150		dBc/Hz
	At 100 kHz		-160.5		dBc/Hz
	At 1 MHz		-161		dBc/Hz
Supply current of each clock	Measured from VRFCK1		1.3		mA

Table 2-17. RF Clock output buffer specifications (RFCK2A, RFCK2B, 52 MHz)

Test conditions: TSX = 52 MHz, 20 pF capacitive load for each buffer output

Parameter	Condition	Min.	Typical	Max.	Unit
Output frequency			52		MHz
Output swing	Set by VRFCK2	1.35	1.4	1.45	V
Duty cycle		47		53	%
Output impedance	1X	40	50	60	Ω
	2X	25	30	35	Ω
	3X	20	25	30	Ω
	4X	12	15	18	Ω
Phase noise	At 10 Hz		-77		dBc/Hz
	At 100 Hz		-107		dBc/Hz
	At 1 kHz		-136		dBc/Hz
	At 10 kHz		-153		dBc/Hz
	At 100 kHz		-159.5		dBc/Hz
	At 1 MHz		-160		dBc/Hz
Supply current of each clock	Measured from VRFCK2		2.2		mA

Table 2-18. RF Clock output buffer specifications (RFCK2A, RFCK2B, 26 MHz)

Test conditions: TSX = 26 MHz, 20 pF capacitive load for each buffer output

Parameter	Condition	Min.	Typical	Max.	Unit
Output frequency			26		MHz
Output swing	Set by VRFCK2	1.35	1.4	1.45	V

Parameter	Condition	Min.	Typical	Max.	Unit
Duty cycle		47		53	%
Output impedance	1X	40	50	60	Ω
	2X	25	30	35	Ω
	3X	20	25	30	Ω
	4X	12	15	18	Ω
Phase noise	At 10 Hz		-80		dBc/Hz
	At 100 Hz		-110		dBc/Hz
	At 1 kHz		-142.5		dBc/Hz
	At 10 kHz		-150.5		dBc/Hz
	At 100 kHz		-157.5		dBc/Hz
	At 1 MHz		-158		dBc/Hz
Supply current of each clock	Measured from VRFC2		1.3		mA

RFCK2 measurement results:

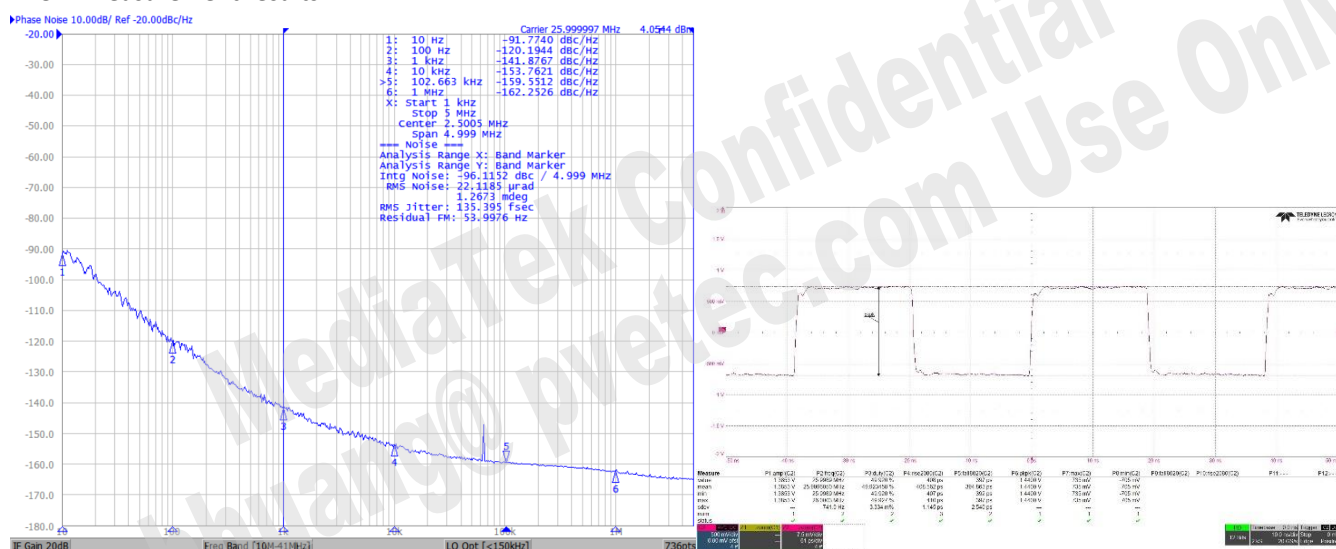
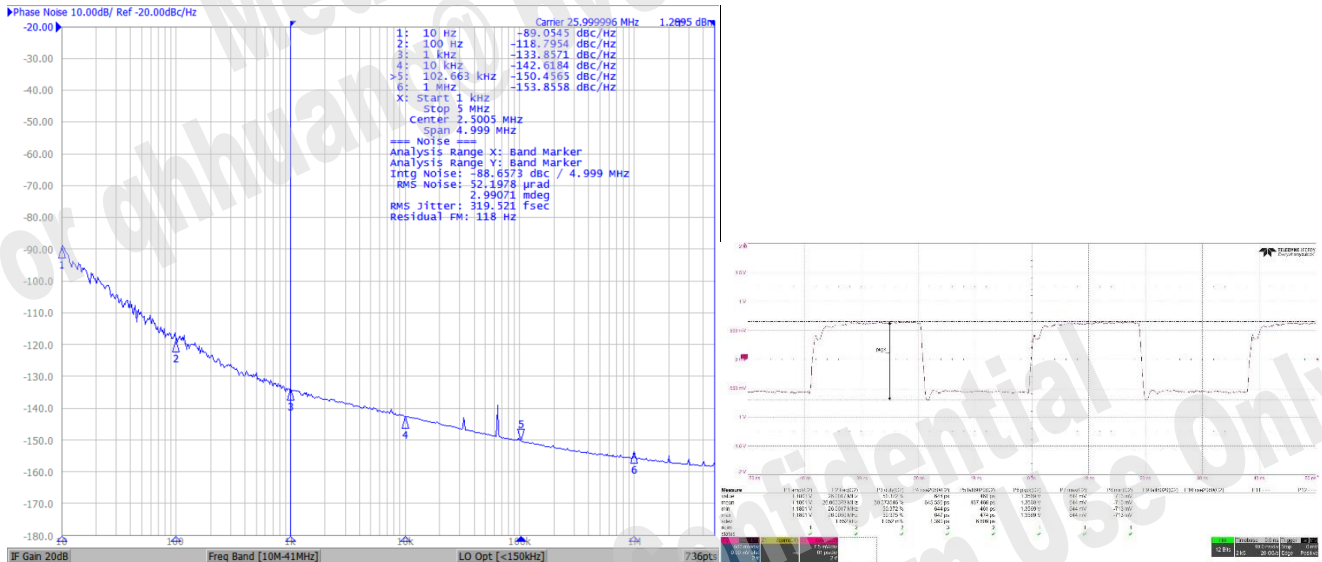


Table 2-19. BB clock output buffer specifications (BBCK1~5, 26 MHz)

Parameter	Condition	Min.	Typical	Max.	Unit
Output frequency			26		MHz
Output swing	Set by VBCK	1.15	1.2	1.25	V
Duty cycle		45		55	%
Output impedance	1X	50	60	70	Ω
	2X	45	55	65	Ω
	3X	40	50	60	Ω
	4X	30	40	50	Ω
Phase noise	At 10 Hz		-80		dBc/Hz
	At 100 Hz		-110		dBc/Hz
	At 1 kHz		-130		dBc/Hz
	At 10 kHz		-140		dBc/Hz
	At 100 kHz		-150		dBc/Hz

Parameter	Condition	Min.	Typical	Max.	Unit
	At 1 MHz		-150		dBc/Hz
RMS jitter	Integrated BW: 1k ~ 13 MHz		2		ps
Supply current of each clock	Measured from VBBCK		1		mA

BBCK measurement results:



3 Functional Description

3.1 General Description

MT6685 is a fully integrated standalone DCXO target for 5G-Module clock provider. The following is the block diagram of MT6685 clock IC.

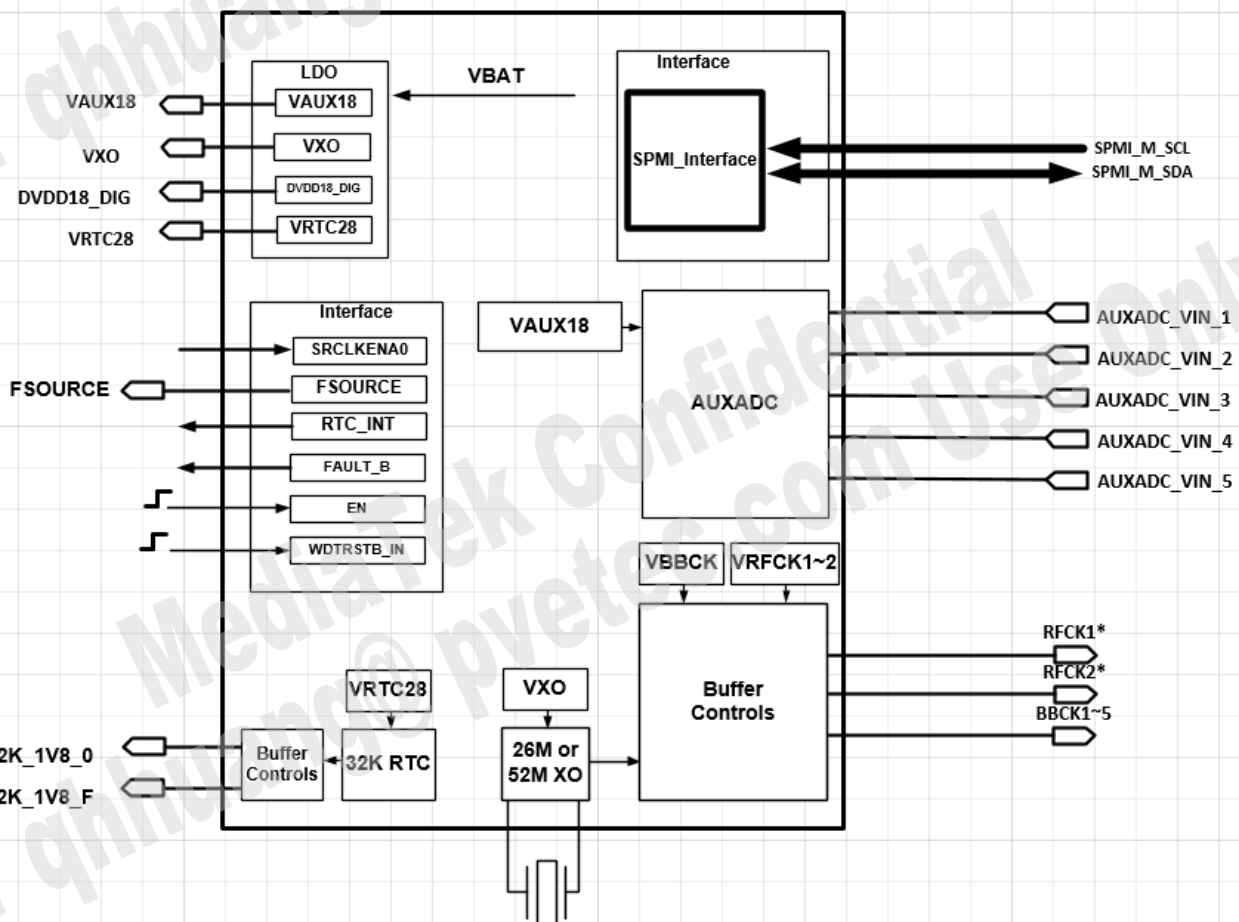


Figure 3-1. MT6685 block diagram

3.2 Clock IC Functional Blocks

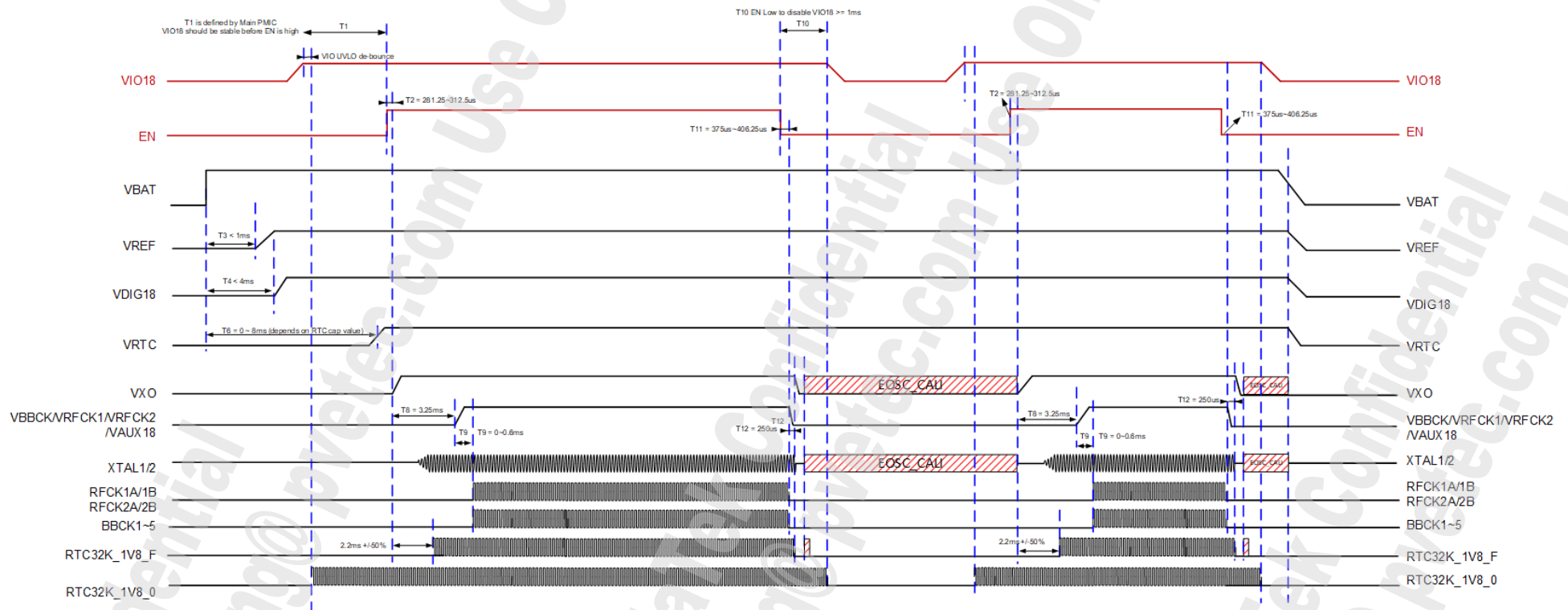
MT6685 manages the clock supply of 5G-Module and other portable systems. MT6685 includes the following analog functions.

- RFCK and BBCK: Provides low noise and low power clock output
- ADC: Provide 15-bit AUXADC for thermal/accessory detection monitor and measurement
- Controller: Generates power-on/off sequence, system reset and exceptional handling function

More detailed descriptions of each sub-block are explained in the following sections.

3.2.1 Power-on/off Sequence

This clock IC handles the power-on and power-off of the system. If the VSYS voltage is neither in the UVLO state nor in the thermal condition, there is only one method to power on the system.



Notes

1. The timings are typical values; timing variation for T3/T4 is $\pm 50\%$, for others are $\pm 10\%$.
2. The start-up time of VRTC is on the condition of output R/C. It takes longer if you use output cap only for VRTC output for BOM reduction.
3. T11 and T12 are the normal off delay time with setting RG_SEQ_OFF = 1'b1 before system power off
4. VBAT, VREF, VDIG18, and VRTC ready time should follow the sequence but don't care about the actual time. They should be ready before EN goes high.

Figure 3-2. MT6685HP/B power-on/off control sequence without XTAL by EN high

EN high (pulling the EN pin to high level)

Pulling EN high is a typical method to turn on the system. The system reset ends at the moment when all default-on regulators are sequentially turned on.

Under-voltage lockout (UVLO)

The UVLO state in the clock IC prevents start-up if the initial voltage of the VSYS is below UVLO_VTH. The judgment is done by VSYSNS. It ensures that the system is powered on with the VSYS in good condition. The UVLO function is performed by a hysteretic comparator which ensures smooth power-on sequence. In addition, when the VSYS voltage is getting lower, it enters the UVLO state and the clock IC is turned off by itself, except for VRTC, to prevent further discharging. Once the clock IC enters the UVLO state, it draws low quiescent current. VRTC still works until DDLO disables it.

Deep discharge lockout (DDLO)

The clock IC enters the deep discharge lockout (DDLO) state when the VSYS voltage drops below DDLO_VTH. In this state, VRTC is shut down. Therefore, it draws very low quiescent current to prevent further discharging or even damage to the cells.

3.2.2 Low Dropout Regulator (LDOs) and Application Reference

Table 3-1. LDO types and brief specifications

LDO name	Controller power domain	Input power domain	Default voltage (V)	Output voltage (V)	I _{max} (mA)	Application
VBCK	DVDD18_IO	DVDD18_IO	ON (1.2)	1.2	10	CLOCK IC internal use (DCXO)
VRFC1/VRFC2	DVDD18_IO	DVDD18_IO	ON (1.4)	1.4	10	CLOCK IC internal use (DCXO)
VXO	DVDD18_IO	DVDD18_IO	ON (1.6)	1.6	5	CLOCK IC internal use (DCXO)
VAUX18	VBAT	VBAT	ON (1.84)	1.84	3	CLOCK IC internal use (AUXADC)
VRTC28	VBAT	VBAT	ON (2.8)	2.8	2	CLOCK IC internal use (RTC)
DVDD18_DIG	VBAT	VBAT	ON (1.8)	1.8	10	CLOCK IC internal use (Digital block)

3.2.3 AUXADC

3.2.3.1 Block Description

The auxiliary ADC includes the following functional blocks:

1. Analog multiplexer: Selects signal from one of the ADC inputs. Real-world messages to be monitored, e.g., temperature, should be transferred to the voltage domain.
2. Pull-high resistor: Input voltage is defined by pull-high resistor and thermistor.
3. Input buffer: Isolates external nodes from internal resistance, and drives A/D converter
4. 15-bit A/D converter: Converts the multiplexed input signal to 15-bit digital data.

Table 3-2. Application and input range of XOADC channels

Channel	Application	Input range (V)
0	AUXADC_VIN1-4	0.04~1.78
0	AUXADC_VIN5 or ext. 100-k R	0.04~1.78
Others	Internal use	N/A

3.2.4 Real-Time Clock

The real-time clock (RTC) module provides time and data information. The clock is based on 32 kHz from DCXO or an embedded 32 kHz oscillator. In addition to providing timing data, an alarm interrupt is generated and can be used to power up the baseband core. Regulator interrupts corresponding to seconds, minutes, hours and days can be generated whenever the time counter value reaches a maximum value (e.g., 59 for seconds and minutes, 23 for hours, etc.). The year span up to 2,127 is supported. The maximum day-of-month values, which depend on the leap year condition, are stored in the RTC block.

In MT6685, the RTC module only supports function without 32 kHz crystal. The RTC module has an embedded 32 kHz oscillator to assist life-time extension after DCXO 32 kHz clock is off.

3.2.5 DCXO

MT6685 clock IC integrates a core 26 MHz or 52M XTAL oscillator (XO) and supports 32 kHz RTC-removal low power mode. It also provides backward compatible (BCM) configurations for the previous solutions. There are four high performance (1.4V_{PP}) and five low power (1.2V_{PP}) external clock buffers for the entire system and one internal buffer for audio.

The temperature sensing crystal with thermistor is connected between XTAL1 and XTAL2.

3.2.5.1 32 kHz RTC Removal Mode

In the low power mode, a 32 kHz clock is produced by a fractional-N divider (1st order sigma delta modulated) to replace the traditional 32 kHz XTAL. The capacitance is minimized for significant current reduced.

Parameter	Condition	Min.	Nom.	Max.	Unit
RTC32K_1V8_0 frequency	Averaged value		32768		Hz
RTC32K_1V8_F frequency	Averaged value		26M/800 ^(Note)		Hz
XO LPM current consumption	Low power mode With all capacitors switched off and auto-amplitude calibration on		60		μA
Duty cycle		45		55	%
Settling time				5	ms
Frequency variation	After system calibration		Same as DCXO		ppm
	Over temperature -30°C to 85°C			15	ppm
	Uncalibrated - total variation			200	ppm

Note: Reserve XO_DIG32K_FIXCK_SEL (default 1'b0) to select /800 (1'b0) or /792 (1'b1) 32K clock

3.2.6 Interrupt and Watchdog

3.2.6.1 Interrupt

See below for the interrupt list of the clock IC to inform BB IC:

1. LDO OC interrupt
The clock IC supports LDO OC interrupt generation. It is issued if any one of the LDOs has OC condition.
2. RTC interrupt

3.2.6.2 Watchdog Reset

WDTRSTB_IN is the watchdog reset from AP. The clock IC resets all modules to the initial state when receiving watchdog reset from AP.

3.2.6.3 Booting Watchdog Reset

The clock IC implements a booting watchdog function to prevent AP from not booting up correctly. After all default on powers are turned on, WDTRSTB_IN should be set high in a pre-determined time period, or the clock IC shuts down and resets all modules to the initial state if EN goes low. All powers are re-powered on again if the power-on condition still exists.

3.2.7 SPMI Interface

MT6685 uses a two-wire MIPI SPMI interface consisting of one clock and one data (SDATA and SCLK) to connect to the main chip. This interface allows the main chip to write commands to and read status from the clock IC.

The SPMI slave provides access to the programmable functions and registers on the device. This protocol uses a two-wire interface for bi-directional communications between ICs connected to the bus. The two interface lines are the serial data line (SDATA) and the serial clock line (SCLK). Every device on the bus is assigned a unique address and acts as either a master or slave depending on whether it generates or receives the serial clock SCLK. The SCLK and SDATA lines should each have a pull-down resistor placed somewhere on the line and remain LOW even when the bus is idle. Note that the SCLK pin is not used for serial bus data transfer. MT6685 supports SPMI speed up to 26 MHz.

- Access register in clock IC by MIPI SPMI protocol
- Supports MIPI SPMI v2.0 read/write command full set (9 commands).
- Supports MIPI SPMI v2.0 four power related command sets (reset/sleep/shutdown and wakeup).
- Supports MIPI SPMI v2.0 RCS feature, which allows SPMI slave to send write commands with interrupt information to SPMI master and SPMI slave if the master supports RCS function.

MT6685 MIPI SPMI slave supports the following formats.

- Power related command
 - Reset
 - Sleep
 - Shutdown
 - Wakeup
- Register read/write command
 - Register 0 write
 - Register read/write
 - Extended register read/write
 - Extended register read/write long
- Slave issued RCS command
 - Master write
 - Extended register write

3.2.8 GPIO

3.2.8.1 GPIO List

Table 3-3. MT6685 GPIO list

Pin Name	Aux Func.0	Aux Func.1	Aux Func.2	Aux Func.3	Aux Func.4	Aux Func.5	Aux Func.6	Aux Func.7	PU/PD	PullEn	PullSel	Mode
PAD_SRCLKENA0	GPIO0	I1:SRCLKENA0				I0:TEST_CK0	I0:TEST_IN0	O:TEST_OUT0	PU/PD	1	1	1
PAD_RTC32K_1V8_	GPIO1	O:RTC32K_1V8_0					I0:TEST_IN1	O:TEST_OUT1	PU/PD	0		1
PAD_RTC32K_1V8_	GPIO2	O:RTC32K_1V8_F					I0:TEST_IN2	O:TEST_OUT2	PU/PD	0		1
PAD_SPML_SCL	GPIO3	I0:SPMI_SCL							PU/PD	0		1
PAD_SPML_SDA	GPIO4	B0:SPMI_SDA							PU/PD	0		1

3.2.8.2 GPIO Specifications

Table 3-4. MT6685 GPIO electrical characteristics

Parameter	Description	Min.	Typical	Max.	Unit
Input					
VIH	Input logic high voltage	0.75*VIO		VIO + 0.3	V
VIL	Input logic low voltage	-0.3		0.25*VIO	V
Rpu	Input pull-up resistance	40	75	190	KΩ
Rpd	Input pull-down resistance	40	75	190	KΩ
Output					
VOH (DC)	DC output logic high voltage	0.75*VIO			V
VOL (DC)	DC output logic low voltage			0.25*VIO	V
Leakage					
IIN	Input leakage current (any input 0V < VIN < VDDIO)	-5		5	uA
IOZ	Tri-state output leakage current	-5		5	uA
IIN	Input leakage current (VIN = 3.3V/0V) for floating nwell I/O	-10		10	uA
IOZ	Tri-state output leakage current for floating nwell I/O	-10		10	uA

3.3 Register Table and Description

Module name: MT6685_PMIC_Register_Mapping_(E1) Base address: (+0h)

Address	Name	Width	Register Function
0000000C	<u>PONSTS</u>	8	Power on Source Record Register
0000000D	<u>POFFSTS0</u>	8	Power off Source Record Register 0
0000000E	<u>POFFSTS1</u>	8	Power off Source Record Register 1
00000010	<u>PSTSCTL</u>	8	Power on/off Status Control
00000013	<u>PG_SDN_STS0</u>	8	Power Good Shutdown Status Register 0
00000014	<u>OC_SDN_STS0</u>	8	OC Shutdown Status Register 0
00000127	<u>TOP_RST_MISC</u>	8	Reset Control Misc
0000079A	<u>DCXO_EXTBUF1_CW0</u>	8	DCXO External Buffer Control 1
0000079B	<u>DCXO_EXTBUF2_CW0</u>	8	DCXO External Buffer Control 2
0000079C	<u>DCXO_EXTBUF3_CW0</u>	8	DCXO External Buffer Control 3
0000079D	<u>DCXO_EXTBUF4_CW0</u>	8	DCXO External Buffer Control 4
0000079E	<u>DCXO_EXTBUF5_CW0</u>	8	DCXO External Buffer Control 5
0000079F	<u>DCXO_EXTBUF6_CW0</u>	8	DCXO External Buffer Control 6
000007A0	<u>DCXO_EXTBUF7_CW0</u>	8	DCXO External Buffer Control 7
000007A1	<u>DCXO_EXTBUF8_CW0</u>	8	DCXO External Buffer Control 8
000007A2	<u>DCXO_EXTBUF9_CW0</u>	8	DCXO External Buffer Control 9
000007A3	<u>DCXO_EXTBUF10_CW0</u>	8	DCXO External Buffer Control 10
000007BC	<u>DCXO_32KDIV_CW1_L</u>	8	DCXO 32KDIV Control Code Word 1 [7:0]
000007BD	<u>DCXO_32KDIV_CW1_H</u>	8	DCXO 32KDIV Control Code Word 1 [15:8]
000007BE	<u>DCXO_32KDIV_CW2_L</u>	8	DCXO 32KDIV Control Code Word 2 [7:0]
000007BF	<u>DCXO_32KDIV_CW2_H</u>	8	DCXO 32KDIV Control Code Word 2 [15:8]
000007C0	<u>DCXO_CDAC_CW1</u>	8	DCXO CDAC Control Code Word 1
000007C2	<u>DCXO_AAC_CW1</u>	8	DCXO AAC Control Code Word 1

Address	Name	Width	Register Function
000007F0	<u>DCXO AAC ELR 0</u>	8	DCXO AAC ELR 0 Register
00000A0C	<u>STRUP ANA CON4</u>	8	STRUP Control Register 4
00001088	<u>SDMADC ADC0_L</u>	8	SDMADC ADC Register 0 L Byte
00001089	<u>SDMADC ADC0_H</u>	8	SDMADC ADC Register 0 H Byte
0000108A	<u>SDMADC_STA0</u>	8	SDMADC_STA0

0000000C **PONSTS** **Power on Source Record Register** **00000001**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																STS_EN ON
Type																RO
Reset																1

Bit(s)	Name	Description
0	STS_ENON	Power on for EN press

0000000D **POFFSTS0** **Power off Source Record Register 0** **00000000**

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									STS_PU PSRC				STS_PS OC	STS_PG FAIL	STS_UV LO_VSYS	STS_DD LO
Type									RO				RO	RO	RO	RO
Reset									0				0	0	0	0

Bit(s)	Name	Description
7	STS_PUPSRC	Power off for power on source missing
3	STS_PSOC	Power off for default on regulator OC
2	STS_PGFAIL	Power off for PWRGOOD failure
1	STS_UVLO_VSYS	Power off for UVLO VSYS event
0	STS_DDLO	Power off for DDLO event

0000000E POFFSTS1 Power off Source Record Register 1 00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name													STS_UV LO_VIO 18	STS_EN OFF_SE Q	STS_EN OFF	
Type													RO	RO	RO	
Reset													0	0	0	

Bit(s)	Name	Description
3	STS_UVLO_VIO18	Power off for UVLO VIO18 event
2	STS_ENOFF_SEQ	Power off for EN sequence off
1	STS_ENOFF	Power off for EN immediately off

00000010 PSTSCTL Power on/off Status Control 00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name															RG_PO NSTS_C LR	RG_POF FSTS_CL R
Type															RW	RW
Reset															0	0

Bit(s)	Name	Description
1	RG_PONSTS_CLR	Clears PONSTS
0	RG_POFFSTS_CLR	Clears POFFSTST and PG/OC status

00000013 PG_SDN_STS0 Power Good Shutdown Status Register 0 0000007F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name										STRUP_ VAUX18 _PG_ST ATUS		STRUP_ VRFCK2 _PG_ST ATUS	STRUP_ VRFCK1 _PG_ST ATUS	STRUP_ VBBCK _PG_STA TUS	STRUP_ VRTC28 _PG_ST ATUS	
Type										RO		RO	RO	RO	RO	
Reset										1		1	1	1	1	

Bit(s)	Name	Description
6	STRUP_VAUX18_PG_STATUS	Shutdown PG status (cleared by software) 0: Power not good

Bit(s)	Name	Description
4	STRUP_VRFCK2_PG_STATUS	1: Power good Shutdown PG status (cleared by software) 0: Power not good
3	STRUP_VRFCK1_PG_STATUS	1: Power good Shutdown PG status (cleared by software) 0: Power not good
2	STRUP_VBBCK_PG_STATUS	1: Power good Shutdown PG status (cleared by software) 0: Power not good
1	STRUP_VRTC28_PG_STATUS	1: Power good Shutdown PG status (cleared by software) 0: Power not good

00000014				OC SDN_STS0				OC Shutdown Status Register 0								00000000			
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16			
Name																			
Type																			
Reset																			
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0			
Name										STRUP_VAUX18_OC_STATUS		STRUP_VRFCK2_OC_STATUS	STRUP_VRFCK1_OC_STATUS	STRUP_VBBCK_OC_STATUS	STRUP_VRTC28_OC_STATUS	STRUP_VXO_OC_STATUS			
Type										RO		RO	RO	RO	RO	RO			
Reset										0		0	0	0	0	0			

Bit(s)	Name	Description
6	STRUP_VAUX18_OC_STATUS	Shutdown OC status (cleared by software) 0: No OC 1: OC occurs.
4	STRUP_VRFCK2_OC_STATUS	Shutdown OC status (cleared by software) 0: No OC 1: OC occurs.
3	STRUP_VRFCK1_OC_STATUS	Shutdown OC status (cleared by software) 0: No OC 1: OC occurs.
2	STRUP_VBBCK_OC_STATUS	Shutdown OC status (cleared by software) 0: No OC 1: OC occurs.
1	STRUP_VRTC28_OC_STATUS	Shutdown OC status (cleared by software) 0: No OC 1: OC occurs.
0	STRUP_VXO_OC_STATUS	Shutdown OC status (cleared by software) 0: No OC 1: OC occurs.

00000127 TOP_RST_MISC Reset Control Misc 00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name															RG_WD TRSTB_ DEB	RG_WD TRSTB_ EN
Type															RW	RW
Reset															0	0

Bit(s)	Name	Description
1	RG_WDTRSTB_DEB	Enables WDTRSTB debounce 1'b0: No debounce 1'b1: Add debounce 1.6 ms
0	RG_WDTRSTB_EN	WDTRSTB (external watchdog) reset from AP 1'b0: Disable 1'b1: Enable

0000079A DCXO_EXTBUF1_CW0 DCXO External Buffer Control 1 00000035

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name											RG_XO_EXTBUF_ BBCK1_HD	RG_XO_EXTBUF_BBCK1_ RSEL				
Type											RW	RW				
Reset											0	0	1	1	0	

Bit(s)	Name	Description
7:6	RG_XO_EXTBUF_BBCK1_HD	Adjusts output driving strength XO control signal of BBCK1 output driving strength 00: Max. 01: Large 10: Small 11: Min.
5:3	RG_XO_EXTBUF_BBCK1_RSEL	Selects clock output impedance for impedance match 100: 60 ohm 101: 55 ohm 110: 50 ohm 111: 40 ohm

0000079B DCXO_EXTBUF2_CW0

DCXO External Buffer Control 2

00000035

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									RG_XO_EXTBUF_BBCK2_HD		RG_XO_EXTBUF_BBCK2_RSEL					
Type									RW		RW					
Reset									0	0	1	1	0			

Bit(s)	Name	Description
7:6	RG_XO_EXTBUF_BBCK2_HD	Adjusts output driving strength XO control signal of BBCK2 output driving strength 00: Max. 01: Large 10: Small 11: Min.
5:3	RG_XO_EXTBUF_BBCK2_RSEL	Selects clock output impedance for impedance match 100: 60 ohm 101: 55 ohm 110: 50 ohm 111: 40 ohm

0000079C DCXO_EXTBUF3_CW0

DCXO External Buffer Control 3

00000035

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									RG_XO_EXTBUF_BBCK3_HD		RG_XO_EXTBUF_BBCK3_RSEL					
Type									RW		RW					
Reset									0	0	1	1	0			

Bit(s)	Name	Description
7:6	RG_XO_EXTBUF_BBCK3_HD	Adjusts output driving strength XO control signal of BBCK3 output driving strength 00: Max. 01: Large 10: Small 11: Min.
5:3	RG_XO_EXTBUF_BBCK3_RSEL	Selects clock output impedance for impedance match 100: 60 ohm 101: 55 ohm 110: 50 ohm 111: 40 ohm

0000079D DCXO_EXTBUF4_CW0 DCXO External Buffer Control 4 00000035

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									RG_XO_EXTBUF_BBCK4_HD		RG_XO_EXTBUF_BBCK4_RSEL					
Type									RW		RW					
Reset									0	0	1	1	0			

Bit(s)	Name	Description
7:6	RG_XO_EXTBUF_BBCK4_HD	Adjusts output driving strength XO control signal of BBCK4 output driving strength 00: Max. 01: Large 10: Small 11: Min.
5:3	RG_XO_EXTBUF_BBCK4_RSEL	Selects clock output impedance for impedance match 100: 60 ohm 101: 55 ohm 110: 50 ohm 111: 40 ohm

0000079E DCXO_EXTBUF5_CW0 DCXO External Buffer Control 5 00000035

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									RG_XO_EXTBUF_BBCK5_HD		RG_XO_EXTBUF_BBCK5_RSEL					
Type									RW		RW					
Reset									0	0	1	1	0			

Bit(s)	Name	Description
7:6	RG_XO_EXTBUF_BBCK5_HD	Adjusts output driving strength XO control signal of BBCK5 output driving strength 00: Max. 01: Large 10: Small 11: Min.
5:3	RG_XO_EXTBUF_BBCK5_RSEL	Selects clock output impedance for impedance match 100: 60 ohm 101: 55 ohm 110: 50 ohm 111: 40 ohm

0000079F DCXO_EXTBUF6_CW0 DCXO External Buffer Control 6 00000015

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name											RG_XO_EXTBUF_RFCK1A_RSEL					
Type											RW					
Reset											0	1	0			

Bit(s)	Name	Description
5:3	RG_XO_EXTBUF_RFCK1A_RSEL	Selects clock output impedance for impedance match 000: 50 ohm 001: 30 ohm 010: 25 ohm 011: 15 ohm

000007A0 DCXO_EXTBUF7_CW0 DCXO External Buffer Control 7 00000015

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name											RG_XO_EXTBUF_RFCK1B_RSEL					
Type											RW					
Reset											0	1	0			

Bit(s)	Name	Description
5:3	RG_XO_EXTBUF_RFCK1B_RSEL	Selects clock output impedance for impedance match 000: 50 ohm 001: 30 ohm 010: 25 ohm 011: 15 ohm

000007A1 DCXO_EXTBUF8_CW0 DCXO External Buffer Control 8 00000015

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name											RG_XO_EXTBUF_RFCK1C_RSEL					
Type											RW					
Reset											0	1	0			

Bit(s)	Name	Description
5:3	RG_XO_EXTBUF_RFCK1C_RSEL	Selects clock output impedance for impedance match 000: 50 ohm 001: 30 ohm 010: 25 ohm 011: 15 ohm

000007A2 DCXO_EXTBUF9_CW0 DCXO External Buffer Control 9 00000015

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name											RG_XO_EXTBUF_RFCK2A_RSEL					
Type											RW					
Reset											0	1	0			

Bit(s)	Name	Description
5:3	RG_XO_EXTBUF_RFCK2A_RSEL	Selects clock output impedance for impedance match 000: 50 ohm 001: 30 ohm 010: 25 ohm 011: 15 ohm

000007A3 DCXO_EXTBUF10_CW0 DCXO External Buffer Control 10 00000015

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name											RG_XO_EXTBUF_RFCK2B_RSEL					
Type											RW					
Reset											0	1	0			

Bit(s)	Name	Description
5:3	RG_XO_EXTBUF_RFCK2B_RSEL	Selects clock output impedance for impedance match 000: 50 ohm 001: 30 ohm 010: 25 ohm 011: 15 ohm

000007BC DCXO 32KDIV CW1 L DCXO 32KDIV Control Code Word 1 [7:0] 00000040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									XO_32KDIV_NFRAC_FPM_L							
Type									RW							
Reset									0	1	0	0	0	0	0	0

Bit(s)	Name	Description
7:0	XO_32KDIV_NFRAC_FPM_L	32K divider fractional input during FPM Clock = 26 MHz

000007BD DCXO 32KDIV CW1 H DCXO 32KDIV Control Code Word 1 [15:8] 0000001D

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name										XO_32KDIV_NFRAC_FPM_H						
Type										RW						
Reset										0	0	1	1	1	0	1

Bit(s)	Name	Description
6:0	XO_32KDIV_NFRAC_FPM_H	32K divider fractional input during FPM Clock = 26 MHz

000007BE DCXO 32KDIV CW2 L DCXO 32KDIV Control Code Word 2 [7:0] 00000040

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									XO_32KDIV_NFRAC_LPM_L							
Type									RW							
Reset									0	1	0	0	0	0	0	0

Bit(s)	Name	Description
7:0	XO_32KDIV_NFRAC_LPM_L	32K divider fractional input during LPM Clock = 26 MHz

000007BF DCXO 32KDIV CW2 H DCXO 32KDIV Control Code Word 2 [15:8] 0000001D

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name										XO_32KDIV_NFRAC_LPM_H						
Type										RW						
Reset										0	0	1	1	1	0	1

Bit(s)	Name	Description
6:0	XO_32KDIV_NFRAC_LPM_H	32K divider fractional input during LPM Clock = 26 MHz

000007C0 DCXO CDAC CW1 DCXO CDAC Control Code Word 1 00000088

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name										XO_CDAC_FPM						
Type										RW						
Reset									1	0	0	0	1	0	0	0

Bit(s)	Name	Description
7:0	XO_CDAC_FPM	XO CDAC code (complement) during FPM

000007C2 DCXO AAC CW1 DCXO AAC Control Code Word 1 0000000F

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name										XO_AAC_FPM_SWEN	XO_OSC_ISEL_FPM_MAN	XO_OSC_ISEL_FPM_M				
Type										RW	RW	RW				
Reset										0	0	0	1	1	1	1

Bit(s)	Name	Description
6	XO_AAC_FPM_SWEN	0: Disable 1: Software triggers AAC during FPM.
5	XO_OSC_ISEL_FPM_MAN	XO FPM ISEL enable manual mode
4:0	XO_OSC_ISEL_FPM_M	XO core FPM current settings if XO_AAC_ISEL_MAN = 1

000007F0

DCXO_AAC_ELRL_0

DCXO AAC ELR 0 Register

00000008

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name											RG_XO_AAC_VSEL					
Type											RW					
Reset											0	0	1	0	0	0

Bit(s)	Name	Description
5:0	RG_XO_AAC_VSEL	XO AAC peak-hold Vref if XO_AAC_CMP_MAN = 1 4'b1111: 2.15V 4'b1110: 2.1V 4'b1101: 2.05V 4'b1100: 2.0V 4'b1011: 1.95V 4'b1010: 1.9V 4'b1001: 1.85V 4'b1000: 1.8V 4'b0111: 1.75V 4'b0110: 1.7V 4'b0101: 1.65V 4'b0100: 1.6V 4'b0011: 1.55V 4'b0010: 1.5V 4'b0001: 1.45V 4'b0000: 1.4V

00000A0C

STRUP_ANA_CON4

STRUP Control Register 4

00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name												RG_FAULT_RST	RGS_FAULT			
Type												RW	RO			
Reset												0	0	0	0	

Bit(s)	Name	Description
4	RG_FAULT_RST	FAULT status reset
3:1	RGS_FAULT	FAULT status

00001088 SDMADC_ADC0_L SDMADC ADC Register 0 L Byte 00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									SDMADC_ADC_OUT_CH0_L							
Type									RO							
Reset									0	0	0	0	0	0	0	0

Bit(s)	Name	Description
7:0	SDMADC_ADC_OUT_CH0_L	SDMADC channel 0 output data

00001089 SDMADC_ADC0_H SDMADC ADC Register 0 H Byte 00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name									SDMADC_ADC_RDY_CH0	SDMADC_ADC_OUT_CH0_H						
Type									RO	RO						
Reset									0	0	0	0	0	0	0	0

Bit(s)	Name	Description
7	SDMADC_ADC_RDY_CH0	SDMADC channel 0 output data ready 0: SDMADC data are proceeding. 1: SDMADC data are ready.
6:0	SDMADC_ADC_OUT_CH0_H	SDMADC channel 0 output data

0000108A SDMADC_STA0 SDMADC_STA0 00000000

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Name																
Type																
Reset																
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Name																SDMADC_ADC_BUSY_IN_CH0
Type																RO
Reset																0

Bit(s)	Name	Description
0	SDMADC_ADC_BUSY_IN_CH0	SDMADC CH0 busy status 0: Idle

Bit(s)	Name	Description
		1: Busy

4 Packaging

4.1 Package Dimension

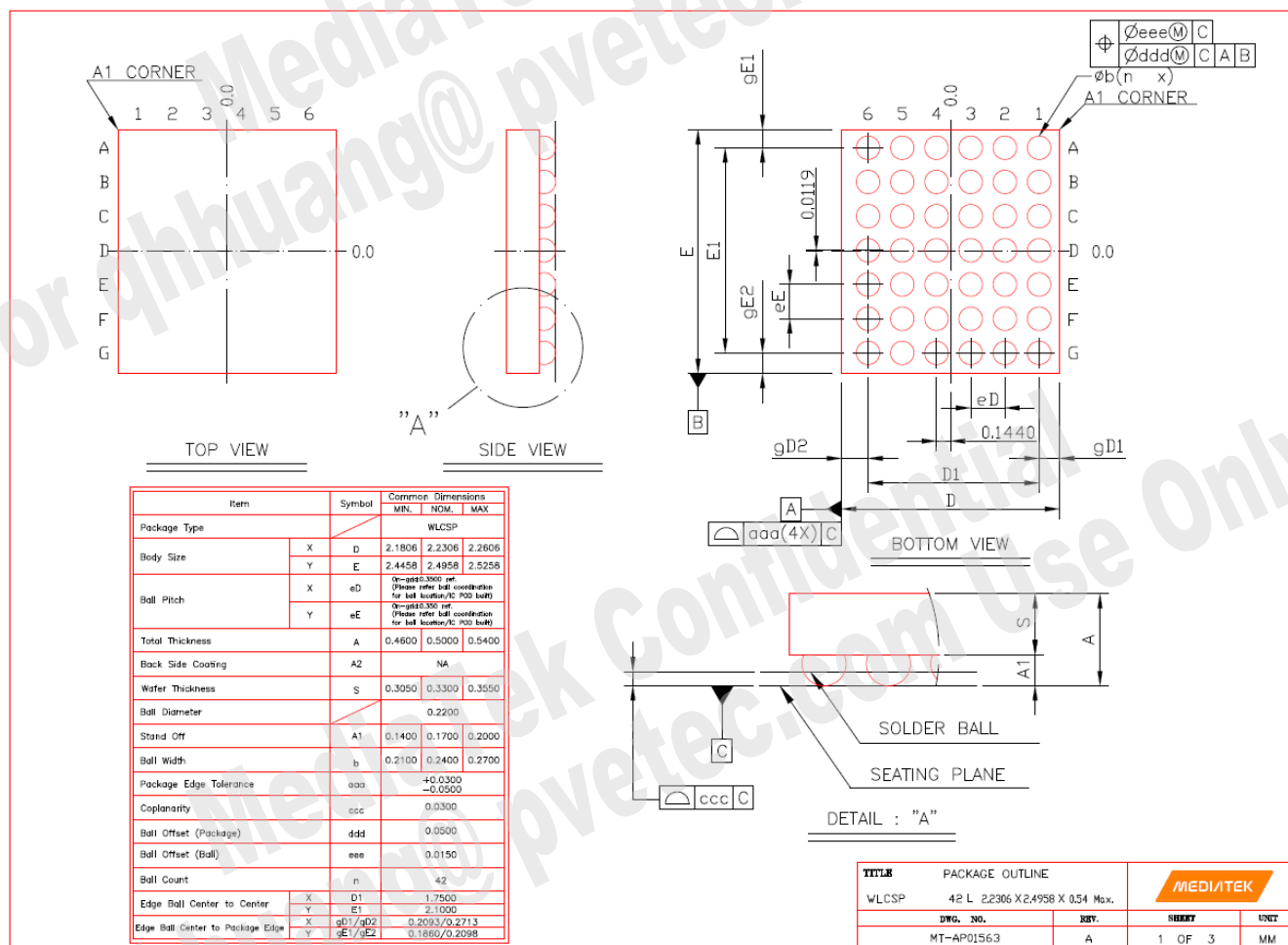


Figure 4-1. Package dimension

Appendix

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